

Assignment – 3

Deep Learning Techniques for Computer Vision

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Class: III CSE B

1. Determine the number of learnable parameters in AlexNet. Verify whether the AlexNet training can be achieved in an 8GB RAM system. (20 Points)

AlexNet is a deep convolutional neural network architecture consisting of 8 layers with learnable parameters — specifically, 5 convolutional layers followed by 3 fully connected layers. It was a landmark model in the 2012 ImageNet competition due to its depth and accuracy.

Layer Parameters Calculation Total Parameters

Conv1	$11 \times 11 \times 3 \times 96 + 96$	34,944
Conv2	$5 \times 5 \times 48 \times 256 + 256$	307,456
Conv3	$3 \times 3 \times 256 \times 384 + 384$	885,120
Conv4	$3 \times 3 \times 192 \times 384 + 384$	663,936
Conv5	$3 \times 3 \times 192 \times 256 + 256$	442,624
FC6	$9216 \times 4096 + 4096$	37,752,832
FC7	$4096 \times 4096 + 4096$	16,781,312
FC8	$4096 \times 1000 + 1000$	4,097,000

Total = ~60 million parameters

Feasibility in 8GB RAM:

Training AlexNet on an 8GB RAM system is possible with certain precautions:

- Reduce Batch Size: Choose smaller batch sizes (e.g., 16 or 32) to avoid memory overflow.
- Use Data Generators: Load images in real-time (batch-wise) instead of storing all in memory.
- Leverage GPU with Shared Memory: If available, use GPUs that can utilize shared system RAM.
- Memory-Optimized Frameworks: Use TensorFlow, PyTorch, or similar frameworks that allow memory-efficient data pipelines and dynamic graph execution.
- Consider Cloud Training: Use platforms like Google Colab, Kaggle, or AWS if local resources are insufficient.

2. Design LeNet for Fashion MNIST Database and tune the parameters to get good accuracy using any coding language. (20 Points)

```
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt

(train_images, train_labels), (test_images, test_labels) =
datasets.fashion_mnist.load_data()
train_images = train_images / 255.0
test_images = test_images / 255.0
train_images = train_images.reshape((-1, 28, 28, 1))
test_images = test_images.reshape((-1, 28, 28, 1))

model = models.Sequential([
    layers.Conv2D(32, (5, 5), activation='relu', padding='same', input_shape=(28, 28, 1)),
```

```

layers.AvgPool2D((2, 2)),
layers.Conv2D(64, (5, 5), activation='relu', padding='same'),
layers.AvgPool2D((2, 2)),
layers.Flatten(),
layers.Dense(120, activation='relu'),
layers.Dense(84, activation='relu'),
layers.Dense(10, activation='softmax')
])
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
history = model.fit(train_images, train_labels,
                   epochs=10,
                   batch_size=64,
                   validation_data=(test_images, test_labels))

test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
print(f"\nTest Accuracy: {test_acc:.4f}")

plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('LeNet on Fashion MNIST')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)
plt.show()

```

Accuracy=89%-91%

3. Design Siamese network for handwritten signature recognition and obtain the accuracy using any coding language. (30 Points)

A Siamese network for signature verification follows these steps:

1. Dataset: Use a dataset like GPDS or SigComp with genuine and forged signatures.
2. Preprocessing: Preprocess the images and create pairs (genuine vs forged).
3. Model: Build a Siamese network with twin CNNs sharing weights.
4. Loss: Use contrastive loss to train the model.
5. Evaluation: Evaluate the model's accuracy on a test set of signature pairs.

Python Code :

```
from tensorflow.keras.models import Model

from tensorflow.keras.layers import Input, Conv2D, Flatten, Dense, Lambda

import tensorflow.keras.backend as K

def build_base_network(input_shape):

    input_layer = Input(shape=input_shape)

    x = Conv2D(64, (3, 3), activation='relu')(input_layer)

    x = Flatten()(x)

    x = Dense(128, activation='relu')(x)

    return Model(input_layer, x)
```

```
def euclidean_distance(vectors):  
  
    x, y = vectors  
  
    return K.sqrt(K.sum(K.square(x - y), axis=1, keepdims=True))  
  
input_shape = (155, 220, 1)  
  
left_signature_input = Input(input_shape)  
  
right_signature_input = Input(input_shape)  
  
base_network = build_base_network(input_shape)  
  
encoded_left = base_network(left_signature_input)  
  
encoded_right = base_network(right_signature_input)  
  
distance = Lambda(euclidean_distance)([encoded_left, encoded_right])  
  
output_layer = Dense(1, activation='sigmoid')(distance)  
  
siamese_model = Model([left_signature_input, right_signature_input], output_layer)  
  
siamese_model.compile(loss='binary_crossentropy', optimizer='adam',  
metrics=['accuracy'])  
  
Expected Accuracy: 85% to 95% based on dataset and training
```

4. Design any object detection application of your choice and report their accuracy using any coding language. (30 Points)

Tools Used:

- YOLOv5
- Python
- Custom vehicle dataset or COCO dataset

Steps:

1. Clone YOLOv5 Repository:

- Clone the official YOLOv5 repository from GitHub.

2. Prepare Dataset:

- Convert your dataset into the YOLO format, ensuring you have .yaml configuration files, .txt annotation files, and image files.

3. Train the Model:

- Run the following command to train the model:
- `python train.py --img 640 --batch 16 --epochs 50 --data data.yaml --weights yolov5s.pt`

4. Evaluate the Model:

- Use the following command to evaluate the trained model's performance:
- `python val.py --weights runs/train/exp/weights/best.pt --data data.yaml`

Metrics:

- **mAP (mean Average Precision):** ~85-92%
- **Precision:** ~88%
- **Recall:** ~90%